Quantum Attention-Enhanced QBiLSTM for Pulmonary Disease Diagnosis

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***Abstract*—Biomedical natural language processing (NLP) stands to benefit significantly from the principles of quantum computing, notably parallelism and superposition, which can enhance pattern recognition in medical texts. This advancement has the potential to improve diagnosis, prognosis, and treatment strategies within precision medicine. Despite these prospects, the practical implementation of deep quantum neural architectures faces challenges such as decoherence, quantum noise, and current hardware limitations, impeding the progress of quantum NLP. In this study, we present the Quantum Attention-Enhanced QBiLSTM model, specifically designed for pulmonary disease classification. We evaluated our model using two datasets: the Indiana University Chest X-ray Collection and the MIMIC-IV X-ray Reports. Our experimental results indicate that our model surpasses existing architectures, including Quantum LSTM, Quantum BiLSTM, and traditional LSTM models. Specifically, on the Indiana University dataset considering the Findings and Impression sections, our model achieved an accuracy of 90.08%, a precision of 90.08%, and an area under the curve (AUC) of 95.44%. On the MIMIC-IV dataset, it attained an accuracy of 83.16%, a precision of 83.23%, and an AUC of 91.25%, demonstrating its robustness across diverse clinical corpora.These findings underscore the potential of quantum-enhanced biomedical NLP in refining clinical decision-making and medical text classification. By integrating quantum attention mechanisms with bidirectional quantum recurrent layers, our approach enhances pattern recognition in radiology reports, paving the way for future advancements in quantum-based healthcare AI applications.**

***Index Terms*—Quantum NLP; Pulmonary disease; Quantum Machine Learning; Quantum Attention Mechanism; Quantum BiLSTM.**

I. Introduction

Quantum computing has recently emerged as a transformative technology with the potential to redefine computational paradigms in biomedical natural language processing (BioNLP). Traditional deep learning models, including transformer-based architectures such as BERT and BioBERT, have demonstrated remarkable success in medical text classification, named entity recognition, and clinical decision support (Lee et al., 2020; Alsentzer et al., 2019) [1], [8]. However, these models face significant challenges, including high computational costs, scalability issues, and inefficiencies in processing complex biomedical texts. As BioNLP tasks require handling vast amounts of unstructured medical data, alternative approaches that leverage quantum computing principles are gaining attention. Quantum-enhanced models offer advantages in parallelism and superposition, enabling more efficient feature extraction and classification in high-dimensional spaces (Schuld & Petruccione, 2018) [10]. Quantum natural language processing (QNLP) is an emerging research area that explores the integration of quantum mechanics with NLP tasks. Early studies have investigated the feasibility of encoding linguistic structures into quantum circuits, demonstrating theoretical advantages in terms of computational efficiency (Coecke et al., 2020) [5]. Variational quantum algorithms, quantum kernel methods, and hybrid quantum-classical models have been proposed to optimize text representation and improve classification accuracy (Biamonte et al., 2017; Havlíček et al., 2019) [3], [6]. Despite these advancements, practical implementation remains constrained by the limitations of current quantum hardware, including decoherence, quantum noise, and qubit connectivity issues (Preskill, 2018) [9]. Researchers have attempted to mitigate these constraints by developing hybrid models that combine classical deep learning techniques with quantum architectures, leading to improved performance in NLP applications (Kerenidis et al., 2020) [7]. In the biomedical domain, quantum-enhanced deep learning models have been applied to clinical text classification, biomedical relation extraction, and medical document retrieval. Quantum-inspired recurrent models, such as Quantum Long Short-Term Memory (QLSTM) and Quantum Bidirectional LSTM (QBiLSTM), have shown promise in capturing long-range dependencies in medical narratives (Bausch, 2020) [2]. These models leverage quantum circuits to encode sequential dependencies more efficiently than classical LSTM networks. Recent studies have explored quantum attention mechanisms to enhance contextual understanding in medical text classification, further improving interpretability and decision-making in clinical applications (Wang et al., 2023) [11]. However, the application of quantum computing in BioNLP remains in its early stages, and several

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challenges must be addressed before achieving widespread adoption. One major limitation is the scarcity of large-scale quantum-ready biomedical datasets, which hinders the training and evaluation of quantum NLP models. Additionally, quantum circuit depth and the limited availability of error-corrected qubits pose constraints on model scalability. Research efforts are ongoing to develop more noise-resilient quantum architectures and optimize hybrid quantum-classical training strategies to overcome these hurdles (Cerezo et al., 2021) [4]. This study builds upon prior work by introducing the Quantum Attention QBiLSTM model, specifically designed for pulmonary disease classification using medical text datasets. By integrating quantum attention mechanisms with bidirectional quantum recurrent layers, the proposed model aims to enhance pattern recognition in radiology reports and improve classification performance. The effectiveness of this approach is evaluated on benchmark datasets, including the Indiana University Chest X-ray Collection and MIMIC-III X-ray Reports, demonstrating its superiority over existing quantum and classical models. This research contributes to the advancement of quantum-enhanced BioNLP, paving the way for more efficient and interpretable AI-driven clinical decision-making.

II. Materials

In our experimentation with the QBiLSTM model, we utilized the Indiana University Chest X-ray Collection [13], a publicly available dataset containing chest radiographs and their corresponding radiology reports. To construct a well-defined corpus for medical text classification, we extracted textual descriptions associated with the radiographic images, specifically focusing on reports that included the Findings section. This subset comprised 1,779 documents, evenly distributed between 890 reports labeled as non-pathological (label 0) and 889 reports associated with pulmonary disease (label 1). Additionally, we created a second group incorporating both the Findings and Impression sections, which resulted in a more extensive dataset of 1,210 non-pathological reports (label 0) and 1,207 pathological reports (label 1). These subsets allowed for a rigorous evaluation of our quantum-enhanced deep learning model, ensuring a diverse representation of radiological findings for lung disease classification.

To assess the generalization capability of our model, we incorporated the MIMIC-IV database, a large-scale, de-identified medical dataset widely used for clinical research. The MIMIC-IV dataset [12] contains 2,321,355 radiology reports from 237,427 patients. For our study, we curated a subset specifically focused on lung cancer diagnosis by filtering reports using ICD-9 diagnostic codes and retaining only those with a structured Findings section. To maintain textual robustness, reports with fewer than 50 words in this section were excluded, yielding a final dataset of 1,304 high-quality reports. This subset was balanced, comprising 699 reports from lung cancer patients and 605 from non-cancer cases. By leveraging MIMIC-IV alongside the Indiana University Chest X-ray Collection, we ensured that our QBiLSTM model was trained and evaluated on diverse datasets, reinforcing its applicability for real-world biomedical NLP tasks. The combination of these two datasets provided a comprehensive evaluation framework, enhancing the model's ability to generalize beyond a single corpus and improving its robustness in clinical text classification.

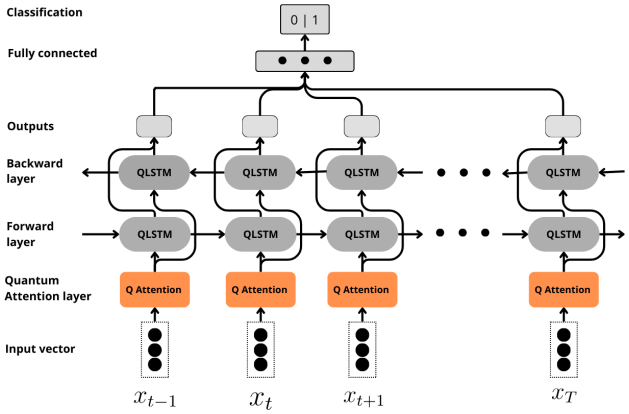


Fig. 1. Proposed Quantum Attention QBiLSTM model Architecture

III. Methods

This study uses a hybrid quantum-classical framework for medical text classification, which includes a Quantum Bidirectional Long Short-Term Memory (QBiLSTM) model to improve sequential learning. The dataset includes of radiological reports with binary labels that have been preprocessed using TF-IDF vectorization and confined to a 512-dimensional feature space to preserve important language patterns. A Quantum Attention Layer (QAL) is crucial to the design, projecting classical embeddings into an 8-qubit Hilbert space using a linear transformation. Angle Embedding encodes the quantum feature representation within a parameterized quantum circuit (PQC) implemented in PennyLane, which includes RY rotations, CNOT gates, and MultiRZ entanglement operations. Expectation values measured in the Pauli-Z basis produce quantum-informed attention ratings, which are then reintegrated into the classical realm.

The QBiLSTM module performs bidirectional processing on quantum-attended representations. Each LSTM gate (forget, input, update, and output) is modeled as a Variational Quantum Circuit (VQC), with TF-IDF embeddings undergoing quantum transformations to optimize hidden state transitions. The final feature representations are converted to binary predictions using a fully connected layer, trained with cross-entropy loss, then optimized with Adam. Figure1 present the Quantum attention Quantum BiLSTM architecture.

A. Quantum Attentiom mechanism

The proposed Quantum Attention Mechanism leverages variational quantum circuits to enhance feature representation by encoding classical data into a quantum state, applying trainable unitary transformations, and capturing dependencies through quantum entanglement. Given an input vector $x \in \mathbb{R}^d$,

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the encoding is performed using Angle Embedding, where each classical feature x_i is mapped onto a qubit state via a rotation along the Y-axis, expressed as:

$$R_Y(x_i) = \begin{bmatrix} \cos(x_i/2) & -\sin(x_i/2) \\ \sin(x_i/2) & \cos(x_i/2) \end{bmatrix} \tag{1}$$

ensuring quantum state preparation. The attention mechanism consists of L parameterized layers, each applying a sequence of quantum operations. First, a set of trainable rotation gates, $R_Y(\theta_i^{(l)})$, is applied to each qubit, where $\theta_i^{(l)}$ are tunable parameters for the lth layer. Next, an entanglement operation is introduced via a sequence of Controlled-NOT (CNOT) gates, defined as:

$$\prod_{i=1}^{n-1} \text{CNOT}_{i,i+1} \tag{2}$$

ensuring that correlations between neighboring qubits are established. The core of the quantum attention mechanism is the application of a MultiRZ gate acting on all qubits simultaneously, defined as:

$$U_{\text{att}}(\phi^{(l)}) = \exp\left(-i\frac{\phi^{(l)}}{2} Z_1 \otimes Z_2 \otimes \cdots \otimes Z_n\right) \tag{3}$$

where $\phi^{(l)}$ is a trainable parameter controlling the global phase interaction among qubits, reinforcing complex dependencies. To ensure bidirectional entanglement propagation, an additional set of reverse CNOT gates:

$$\prod_{i=n-1}^{1} \text{CNOT}_{i,i-1} \tag{4}$$

is applied in all layers except the last. The final quantum state, $|\psi_{\text{out}}\rangle$, is measured in the Pauli-Z basis, where the expectation values:

$$\langle Z_i \rangle = \langle \psi_{\text{out}} | Z_i | \psi_{\text{out}} \rangle \tag{5}$$

are extracted for each qubit, forming a quantum-enhanced feature vector. The complete transformation is expressed as:

$$|\psi_{\text{out}}\rangle = U_{\text{att}}^{(L)} U_{\text{ent}}^{(L)} U_{\text{rot}}^{(L)} \dots U_{\text{att}}^{(1)} U_{\text{ent}}^{(1)} U_{\text{rot}}^{(1)} U_{\text{enc}}(x) |0\rangle^{\otimes n} \tag{6}$$

ensuring that the model effectively captures long-range dependencies in sequential data. This quantum attention mechanism enables a richer, entangled feature space, providing a powerful alternative to classical self-attention models, and holds promise for advancing quantum natural language processing and biomedical text classification.

B. Quantum Bidirectional LSTM (QBiLSTM)

The Quantum Bidirectional Long Short-Term Memory (QBiLSTM) network [15]–[17] extends classical BiLSTM with quantum-enhanced recurrent operations, leveraging Variational Quantum Circuits (VQCs) to encode and process sequential information bidirectionally. Given an input sequence $X = [x_1, x_2, ..., x_T]$, where $x_t \in \mathbb{R}^d$ represents the feature vector at timestep t transformed by Quantum Attention mechanism, the QBiLSTM computes forward and backward hidden representations using Quantum LSTM (QLSTM) cells:

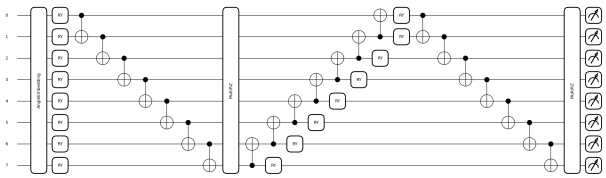


Fig. 2. Variational Quantum Circuit for Quantum Attention mechanism: This figure shows a variational quantum circuit for the attention mechanism with 8 qubits and 2 layers. It encodes inputs using AngleEmbedding, followed by RY rotations, CNOT entanglement gates, and MultiRZ gates for phase adjustments. Reverse CNOT gates enhance entanglement, and final measurements with Pauli-Z operators compute attention scores.

$$\overrightarrow{h}_t, \overrightarrow{c}_t = \text{QLSTM}_f(x_t, \overrightarrow{h}_{t-1}, \overrightarrow{c}_{t-1}) \tag{7}$$

$$\overleftarrow{h}_t, \overleftarrow{c}_t = \text{QLSTM}_b(x_t, \overleftarrow{h}_{t+1}, \overleftarrow{c}_{t+1}) \tag{8}$$

where $\overrightarrow{h}_t$ and $\overleftarrow{h}_t$ represent the forward and backward hidden states, while $\overrightarrow{c}_t$ and $\overleftarrow{c}_t$ denote their respective memory cell states.

1) Quantum LSTM Cell Formulation: Each QLSTM cell [14] consists of four key quantum transformations: the forget gate, the input gate, the update gate, and the output gate, parameterized by trainable quantum circuits. The classical input x_t and previous hidden state h_{t-1} are concatenated:

$$y_t = \text{concat}(h_{t-1}, x_t) \tag{9}$$

This vector is first transformed using a classical linear layer:

$$\tilde{y}_t = W y_t + b \tag{10}$$

where $W \in \mathbb{R}^{q \times (d+h)}$ and $b \in \mathbb{R}^q$. The transformed feature vector $\tilde{y}_t$ is then encoded into a quantum state using angle embedding:

$$|\psi_{\text{enc}}(y_t)\rangle = \bigotimes_{i=1}^{q} R_Y(\tilde{y}_{t,i}) |0\rangle \tag{11}$$

where each classical component $\tilde{y}_{t,i}$ is mapped to a qubit rotation:

$$R_Y(\tilde{y}_{t,i}) = \begin{bmatrix} \cos(\tilde{y}_{t,i}/2) & -\sin(\tilde{y}_{t,i}/2) \\ \sin(\tilde{y}_{t,i}/2) & \cos(\tilde{y}_{t,i}/2) \end{bmatrix} \tag{12}$$

Following encoding, trainable quantum circuits apply entanglement and parameterized transformations. Each of the four QLSTM gates is defined as a quantum variational circuit:

$$U_f(\theta_f) = U_{\text{ent}} U_{\text{rot}}(\theta_f) U_{\text{enc}}(y_t) \tag{13}$$

$$U_i(\theta_i) = U_{\text{ent}} U_{\text{rot}}(\theta_i) U_{\text{enc}}(y_t) \tag{14}$$

$$U_g(\theta_g) = U_{\text{ent}} U_{\text{rot}}(\theta_g) U_{\text{enc}}(y_t) \tag{15}$$

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$$U_o(\theta_o) = U_{\text{ent}} U_{\text{rot}}(\theta_o) U_{\text{enc}}(y_t) \tag{16}$$

where $U_{\text{rot}}(\theta)$ applies trainable single-qubit rotations $R_Y(\theta)$, U_{ent} introduces entanglement via a chain of CNOT gates, $U_{\text{enc}}(y_t)$ encodes the classical input into quantum states. The Pauli-Z expectation values from the output quantum state $|\psi_{\text{out}}\rangle$ are measured to extract quantum-enhanced activations:

$$\langle Z_i \rangle = \langle \psi_{\text{out}} | Z_i | \psi_{\text{out}} \rangle \tag{17}$$

which are then processed through a final classical transformation:

$$f_t = \sigma(W_f \mathbf{z}_f + b_f), \quad i_t = \sigma(W_i \mathbf{z}_i + b_i) \tag{18}$$

$$g_t = \tanh(W_g \mathbf{z}_g + b_g), \quad o_t = \sigma(W_o \mathbf{z}_o + b_o) \tag{19}$$

where $\mathbf{z}_f, \mathbf{z}_i, \mathbf{z}_g, \mathbf{z}_o$ are quantum-enhanced activations retrieved from the quantum circuit outputs. The memory cell state c_t and hidden state h_t are updated as:

$$c_t = f_t \odot c_{t-1} + i_t \odot g_t \tag{20}$$

$$h_t = o_t \odot \tanh(c_t) \tag{21}$$

where $\odot$ represents element-wise multiplication.

2) Final QBiLSTM Representation: The final bidirectional representation is obtained by concatenating the forward and backward hidden states:

$$H = [\overrightarrow{h}_t; \overleftarrow{h}_t] \tag{22}$$

which is then passed through a fully connected layer for classification:

$$\hat{y} = W_{\text{fc}} H + b_{\text{fc}} \tag{23}$$

where $W_{\text{fc}} \in \mathbb{R}^{(2h) \times 2}$ maps the hidden state to binary classification logits.

IV. Results

This section presents the empirical findings derived from our investigation into the application of the Quantum Attention QBiLSTM model for pulmonary disease classification. We systematically evaluated the model's performance across two biomedical datasets, including the Indiana University Chest X-ray Collection and MIMIC-IV, to assess its robustness and generalizability. Key performance metrics accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and Area Under the Curve (AUC) were computed to facilitate a comprehensive comparison with existing architectures such as QBiLSTM, QLSTM, and classical LSTM. The results are organized to highlight the model's efficacy in capturing complex clinical dependencies and its potential to enhance decision-support systems in precision medicine.

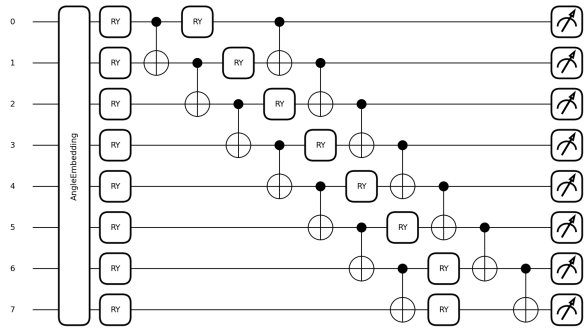


Fig. 3. Variational Quantum Circuit integrated in QBiLSTM architecture: This figure present the variational quantum circuit consisting of 8 qubits and 2 layers. The quantum circuit starts with an input encoding step using the AngleEmbedding, followed by multiple layers of quantum gates. In each layer, rotations (RY gates) are applied to each qubit, and entanglement is introduced using CNOT gates between adjacent qubits. The final state is measured using the Pauli-Z operator for each qubit.

We implemented and evaluated our quantum-enhanced deep learning model in a high-performance computing environment optimized for quantum machine learning. The experiments were conducted using PennyLane version 0.39.0 on a quantum simulator. The setup included an x86_64 architecture with an 8-core processor, 62.50 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU for tensor-based computations. Our Quantum Attention QBiLSTM model, along with QBiLSTM and QLSTM models, used 8 qubits and 2 quantum layers. Models were trained with the Adam optimizer (learning rate 0.001) and preprocessed with a TF-IDF vectorizer (256 features).

A. Comparative results

The experimental results in TableI show the superior performance of the Quantum Attention QBiLSTM model in classifying pulmonary diseases. Tested on the Indiana University Chest X-ray Collection (Findings and Impression), the model achieved the highest precision, recall, F1-score, and accuracy (90.08%), outperforming QBiLSTM, QLSTM, and classical LSTM models. The Matthews Correlation Coefficient (MCC) of 80.19% indicates a well-balanced classification, while the Area Under the Curve (AUC) score of 95.44% reflects strong discriminatory power. On the Indiana University dataset (Findings only), the model still outperformed others, achieving 85.84% precision and 85.77% accuracy. However, the performance drop in this dataset highlights the importance of richer clinical data, as the absence of the Impression section reduced the contextual information. Overall, the quantum attention mechanism shows robust generalization, confirming its potential for real-world clinical applications.

In TableII the results on the MIMIC-IV dataset highlight the superior performance of the Quantum Attention QBiLSTM model, which achieves the highest precision (83.23%), recall (82.46%), F1-score (82.73%), and accuracy (83.16%) among all models. These improvements indicate the effectiveness

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TABLE I
COMPARISON OF EVALUATION METRICS ON INDIANA UNIVERSITY CHEST X-RAY COLLECTION DATASET

	Indiana University dataset (Findings and Impression)				Indiana University dataset (Findings)			
Metric Name	**Proposed Model**	**QBiLSTM**	**QLSTM**	**LSTM**	**Proposed Model**	**QBiLSTM**	**QLSTM**	**LSTM**
Precision	90.08%	89.53%	89.27%	89.26%	85.84%	85.41%	84.12%	84.46%
Recall	90.08%	89.54%	89.25%	89.26%	85.78%	85.40%	83.91%	84.28%
F1-Score	90.08%	89.53%	89.25%	89.26%	85.76%	85.39%	83.87%	84.25%
Accuracy	90.08%	89.53%	89.26%	89.26%	85.77%	85.39%	83.90%	84.27%
MCC	80.19%	79.07%	78.52%	78.52%	71.61%	70.81%	68.03%	68.74%
AUC	95.44%	95.30%	95.20%	95.20%	91.79%	91.59%	90.22%	91.00%

of quantum attention in capturing complex dependencies in medical text. Notably, the MCC score (65.68%) confirms a more balanced classification, outperforming classical LSTM by over 10 percentage points. Additionally, the AUC score (91.25%) demonstrates strong discriminatory power. These findings emphasize the advantage of quantum-enhanced architectures in biomedical NLP, particularly for clinical decision-making tasks. clinical decision-making tasks.

TABLE II
COMPARISON OF EVALUATION METRICS ON MIMIC IV DATASET

Metric name	**Proposed model**	**QBiLSTM**	**QLSTM**	**LSTM**
Precision	83.23%	82.77%	80.89	77.35%
Recall	82.46%	81.88 %	81.28	77.72%
F1-Score	82.73%	82.18%	80.98	77.40 %
Accuracy	83.16%	82.65%	81.12	77.55%
MCC	65.68 %	64.64%	62.17	55.06%
AUC	91.25%	91.32%	90.75	88.23%

The cross-validation results demonstrate the robustness and generalization capability of the Quantum Attention QBiLSTM model across multiple datasets. On the Indiana University Chest X-ray Collection (Findings and Impression) dataset TableIII, the proposed model exhibited superior precision (91.15%), recall (91.88%), and F1-score (91.49%), underscoring its ability to capture intricate clinical dependencies. A Matthews Correlation Coefficient (MCC) of 82.99% further validates its effectiveness in balancing sensitivity and specificity. A similar trend was observed in the Indiana University dataset (Findings only) in TableIV, where the proposed model maintained strong performance with an accuracy of 83.70%, precision of 84.05%, and recall of 83.25%.

Table V presents the cross-validation results on the MIMIC-IV dataset, where the proposed model achieved an average accuracy of 79.30%. The precision (80.80%), recall (80.55%), and F1-score (80.66%) were consistently high, indicating stable classification performance across clinical datasets.

DISCUSSION

The integration of quantum computing principles into biomedical natural language processing (BioNLP) represents a significant leap forward in the field of medical text classification. The proposed Quantum Attention-Enhanced QBiLSTM model demonstrates the potential of quantum-enhanced architectures to outperform classical and existing quantum models in pulmonary disease classification. The results from the Indiana University Chest X-ray Collection and MIMIC-IV datasets highlight the model's ability to capture complex dependencies in medical texts, achieving superior accuracy, precision, recall, and F1-scores compared to traditional LSTM, QLSTM, and QBiLSTM models. This success can be attributed to the quantum attention mechanism, which leverages variational quantum circuits to encode classical data into quantum states, enabling richer feature representations through quantum entanglement and superposition.

One of the key advantages of the proposed model is its ability to handle high-dimensional data more efficiently than classical models. The quantum attention mechanism, with its ability to capture long-range dependencies in sequential data, provides a powerful alternative to classical self-attention models. This is particularly beneficial in the context of medical texts, where the relationships between different parts of a report (e.g., Findings and Impression sections) can be crucial for

TABLE III
PERFORMANCE METRICS OF QUANTUM ATTENTION QBILSTM USING 5-FOLD CROSS-VALIDATION ON INDIANA (FINDINGS AND IMPRESSION)

Fold	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**MCC**
1	92.15%	90.80%	93.80%	92.28%	84.34%
2	91.53%	89.41%	94.21%	91.75%	83.18%
3	90.06%	90.72%	89.21%	89.96%	80.13%
4	92.34%	92.50%	92.12%	92.31%	84.68%
5	91.30%	92.34%	90.04%	91.18%	82.63%
Mean	**91.48%**	**91.15%**	**91.88%**	**91.49%**	**82.99%**
Std	**±0.008**	**±0.011**	**±0.019**	**±0.008**	**±0.016**

TABLE IV
PERFORMANCE METRICS OF QUANTUM ATTENTION QBILSTM USING 5-FOLD CROSS-VALIDATION ON INDIANA (FINDINGS)

Fold	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**MCC**
1	83.71%	86.58%	79.78%	83.04%	67.63%
2	84.83%	84.44%	85.39%	84.92%	69.67%
3	82.58%	82.58%	82.58%	82.58%	65.17%
4	82.30%	84.02%	79.78%	81.84%	64.69%
5	85.07%	82.63%	88.70%	85.56%	70.34%
Mean	**83.70%**	**84.05%**	**83.25%**	**83.59%**	**67.50%**
Std	**±0.011**	**±0.014**	**±0.034**	**±0.014**	**±0.022**

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TABLE V
PERFORMANCE METRICS OF QUANTUM ATTENTION QBILSTM USING 5-FOLD CROSS-VALIDATION ON MIMIC IV DATASET

Fold	Accuracy	Precision	Recall	F1-Score	MCC
1	78.54%	79.58%	80.71%	80.14%	56.82%
2	77.78%	78.47%	80.71%	79.58%	55.24%
3	77.78%	81.06%	76.43%	78.68%	55.62%
4	78.54%	80.00%	80.00%	80.00%	56.86%
5	83.85%	84.89%	84.89%	84.89%	67.54%
Mean	**79.30%**	**80.80%**	**80.55%**	**80.66%**	**58.42%**
Std	**±0.022**	**±0.022**	**±0.026**	**±0.021**	**±0.046**

accurate diagnosis. The model's performance on the MIMIC-IV dataset, which includes a large number of radiology reports, further underscores its robustness and generalizability across diverse clinical datasets.

However, the study also highlights some limitations. The performance of the model drops when the Impression section is excluded from the Indiana University dataset, indicating that richer clinical data enhances the model's ability to capture contextual information. This suggests that while quantum-enhanced models offer significant advantages, they still rely on the quality and completeness of the input data. Additionally, the current implementation is limited by the constraints of quantum hardware, such as decoherence and quantum noise, which can affect the stability and scalability of quantum circuits. Future research should focus on developing more noise-resilient quantum architectures and optimizing hybrid quantum-classical training strategies to overcome these challenges.

The findings of this study align with previous research in quantum machine learning (QML) and quantum natural language processing (QNLP). For instance, Schuld and Petruccione (2018) [10] have demonstrated the potential of quantum computing in supervised learning tasks, while Havlicek et al. (2019) [6] have shown that quantum-enhanced feature spaces can improve classification accuracy. The proposed model builds on these foundations by integrating quantum attention mechanisms with bidirectional recurrent layers, offering a novel approach to medical text classification.

CONCLUSION

The Quantum Attention-Enhanced QBiLSTM model represents a significant advancement in the field of biomedical NLP, particularly in the context of pulmonary disease classification. By leveraging the principles of quantum computing, the model achieves superior performance compared to classical and existing quantum models, demonstrating its potential to enhance clinical decision-making and precision medicine. The quantum attention mechanism, with its ability to capture complex dependencies in medical texts, offers a powerful tool for improving the accuracy and interpretability of AI-driven diagnostic systems.

However, the study also highlights the need for further research to address the limitations of current quantum hardware and to explore the integration of quantum-enhanced models with other advanced NLP techniques, such as transformer-based architectures. As quantum computing technology continues to evolve, the potential for quantum-enhanced BioNLP models to revolutionize medical text analysis and clinical decision support systems becomes increasingly promising. Future work should focus on expanding the application of quantum-enhanced models to other biomedical NLP tasks, such as named entity recognition, relation extraction, and medical document retrieval, to further validate their effectiveness and generalizability.

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